MILITARY SPECIFICATION

MAGNESIUM POWDER
(FOR USE IN AMMUNITION)

This Amendment forms a part of Military Specification MIL-M-382C(AR), dated 10 August 1978, and is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

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3.2.4 Change "Table II" to "Table IIb".

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3.4 Add the following two sentences and then Table IIa to the end of the paragraph:

"For Type II magnesium of nominal granulation 16, mesh size 200/325, the particle size distribution shall be determined by Sedigraph 5000D or equivalent as described in section 4.5.11.2. The percentages lying in particle ranges shall be in accordance with Table IIa.

TABLE IIa. Particle Ranges

Particle Ranges, microns		Percent	
Greater than or equal to	But less than	Min	Max
0	4	0	0
4	14	8.4	14.7
14	30	21.6	27.5
30	74	58.3	66.6
74	-	0	4.0

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Table II. Granulation: Change table number "Table II" to "Table IIb".

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Potassium hydroxide solution, technical is also available commercially in

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3.5 Table III. Apparent density: Add the following sentence to the end of the paragraph:

"For Type II, granulation 16, apparent density shall be 0.62 min."

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4.5.11 Delete in its entirety and substitute the following:

"4.5.11 Granulation.

4.5.11.1 Sieve method. Place a weighed portion of approximately 5 g of the sample on the top sieve of a nest of sieves assembled as specified in Table IIb and provided with a bottom pan. Cover and shake for 30 minutes in a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the portions retained by each sieve and calculate to a percentage basis as required.

4.5.11.2 Sedigraph method. The particle size analysis for Type II granulation 16 magnesium shall be determined by the following procedure. This method utilizes the Sedigraph 5000D by Micromeritics Corporation, or equivalent, with graph paper having the Y axis divided into 0.05 inch, each division being 1%, and the X axis being a 3 log cycle scale, each cycle 3 5/16 inches in length. Calculate the rate settings as follows:

$$\text{Rate} = \frac{211.80\ (P-P_o)}{(50/D_m)^2 n} \times \frac{60}{f}$$

Where:

f = power line frequency (usually 60 hertz)

Dm = starting diameter = 100 microns

P = density of magnesium = 1.74 g/cm

P_o = density of 2 - propanol, temperature dependent, see following table.

n = viscosity of 2 - propanol, temperature dependent, see following table.

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TABLE V

Carrier Densities, Viscosities, and Rates as a Function of Temperature

Temp. (°C)	P_o	n	Rate
30	.7769	1.78	459
31	.7761	1.73	472
32	.7752	1.68	486
33	.7744	1.63	502
34	.7736	1.58	518
35	.7728	1.54	532

Clean the instrument according to manufacturer's instructions, and allow it to warm up for at least one half hour. Mechanically tumble the samples of magnesium from Section 4.4.2.1.2 for 5 minutes to insure that the sample to be tested is uniformly representative. Remove one gram of sample from the container and place in 25 ml of 2 - propanol in the beaker used to hold the sample in the instrument. Place the sample in the beaker into a sonic bath for three minutes to insure that magnesium is thoroughly wetted. Contamination of the sample shall be avoided.

4.5.11.2.1 Calibration. In another 30 ml beaker place 25 ml of the same carrier liquid being used for the sample measurement, in this case, 2-propanol. This is the reference solution to be used for setting the zero point on the chart (calibration).

Place the beaker containing the reference solution in the Sedigraph, setting the beaker with the sample solution towards the rear of the sample compartment. This will enable thermal equilibration of the sample to the compartment conditions during calibration, thus avoiding additional waiting later. Carefully put the pump tubing connecting to the cell into the reference

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solution and turn pump on. Allow the liquid to flow through the cell, then examine the cell and tap gently to eliminate any air bubbles, as they may cause erroneous readings. Set the chart pen at zero with the "zero" dial, lock the dial into position, and let the system equilibrate for several minutes.

Press the metal "zero" button below the "zero" dial to bring the pen back to the zero mark. Observe the pen and adjust for any drift using the "zero percent" dial. (A detailed description of the instrument controls is given in the manufacturer's instruction manual). When calibration is completed, turn cell upside down to allow reference solution to drain from the cell.

4.5.11.2.2 Sample analysis. Place the beaker containing the sample in the Sedigraph. Place a magnetic stir bar in the beaker, close the chamber door, and turn on the stirrer. Then insert pump tubing into sample solution and turn on the pump, allowing mixing to continue. Allow solution to flow through the cell, then examine the cell and tap gently to eliminate any air bubbles. Then set the 100% mark with the "100% percent" dial. Set the exact starting diameter at 100 microns. The analysis is complete when the run reaches baseline or the trace becomes asymptotic with the X axis.

NOTE: The instrument door should remain closed as much as possible during the analysis. Additional instructions may be found in the manufacturer's instruction manual. Copies of the manual may be obtained by contacting Micromeritics Instrument Corporation, 5680 Goshen Springs Road, Norcross, Georgia 30093.

Repeat the analysis on two more samples. Take the average of the three analyses and report the data and calculations as follows:

a. Particle size distribution over the range of 100 to 0.36 microns is automatically plotted by the instrument. For a given range there are three results,

where:

X_l = lowest percent value
X_m = middle percent value
X_h = highest percent value

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The set of three results shall be rejected according to Dixon's Range Test if:

$$\frac{X_m - X_l}{X_h - X_l} > 0.941$$

and three more analyses shall be obtained.

b. Record the weight mean diameter, in microns, found at the 50 percent intercept line.

c. Percentages of magnesium in the particle size range may be calculated as follows:

$$\% PS_U - \% PS_L$$

Where:

PS_U = upper particle size value

PS_L = lower particle size value "

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